Healthcare Technology: History, System Selection, and Implementation

Before computer technology was introduced to the health information (HI) profession, every task was completed manually. After arriving in the HI department, the paper chart was assembled into the correct order. Next, it was analyzed to check that all required documentation was present, including signatures. Then the coding professional reviewed the chart and assigned the codes using only a paper codebook. When completed, the chart would be physically filed, taking up inches to feet of shelf space, depending on the length of the visit. At that time, access to the paper chart was limited to one person at a time, in one location. Now, with the implementation of technology, this process has changed immensely!

Health Technology History

HI professionals use computers to facilitate the ease of entering, analyzing, using, and maintaining data. Advances in the types and capabilities of available computer systems have significantly impacted the way health information and related data are shared and communicated. What once were laborious and repetitive tasks can now be accomplished fairly simply without the need to duplicate work, thus increasing productivity and decreasing the possibility of errors.

The use of information systems has impacted many areas of healthcare, including the following:

Patient care

Finance and reimbursement

Public health and epidemiology

Research

Education

Patient safety

Quality of patient care

Access to information

Privacy and security

Due to the reliance on technology, it is important that HI be involved in the selection and implementation of any technology that involves or impacts the patient's health record, especially the electronic health record (EHR) for the healthcare organization. The EHR contains patient information that was formally stored on paper and filed in folders on the shelves. Despite changes in the storage format of patient information from paper to electronic, responsibility of the oversight and maintenance has NOT shifted to Information Technology (IT) but remains with HI.

Information Systems impact patient care in many ways. Although it does not reduce the cost of patient co-pays or allow for longer lengths of stay for inpatients, it has many positive impacts including:

Patient engagement in healthcare

Patient safety

Creation of evidence-based medicine

Healthcare Technology System Selection, Implementation and Maintenance

As technology advances and computers become more and more robust, the size of the computer keeps getting smaller and smaller. One of the first “personal computers” was so large that it would fill an entire room. Only a few decades later, many of us are walking around with a computer in our pocket or purse (that is, a smart phone). While the exterior portions (hardware) of computers are shrinking in size, the interior components (software) are becoming more complex and diversified.

Project management skills are important to have in the HI profession because there are always new projects like implementing a new electronic health record or coding system. HI professionals have the background and skill set needed to be critical, contributing members of a technology system selection and implementation team. Having project management skills will enable the HI professional to lead as well as participate in rolling out and maintaining new systems efficiently and effectively.

System Uses and Interoperability

There are many different types of technology and information systems utilized throughout the healthcare industry. Some of these systems are used by health information (HI) professionals in their daily job functions, requiring knowledge and understanding of health information and how it integrates with technology. Although other systems that involve patient health information may be primarily used by different departments, HI professionals are still involved with the selection and implementation as well as the management of healthcare information.

Administrative and Clinical Information Systems

Administrative and clinical information systems are found in every healthcare organization. HI staff are unique as they are involved in both types that relate to patient health information. Learn about these two types of systems:

Administrative

Administrative information systems are used to manage the business of healthcare. These systems include the master patient index (MPI), decision support system, registration and admitting, the hospital information system, and the financial system. Many of the users of administrative information systems are not in direct patient care, but their jobs are vital to the business of healthcare and assuring the facility is sustained for those treating the patients.

Clinical Information Systems

Healthcare organizations do not use one central information system to document clinical information. In fact, there are often different systems that are specific to a department (i.e. the radiology department). Radiology may have a separate system to digitally capture and store images. The different systems may be maintained by different vendors, but healthcare organizations strive to make the sharing of patient data centralized through the use of the electronic health record (EHR). The goal is to capture complete and accurate data in the various clinical information systems and use integration and interfaces to the EHR whenever needed.

Care and Maintenance of Chargemasters

"The chargemaster-also called the charge description master (CDM)-is simply a master price list of supplies, devices, medications, services, procedures, and other items for which a distinct charge to the patient exists. It is a financial management form that contains information about the organization’s charges for the healthcare services it provides to patients."

As mentioned above, the **charge description master**, also called the **chargemaster** or **CDM**, is a list of supplies and services available in a healthcare facility. Each line-item charge includes some key elements including:

- Charge description
- Charge amount
- CPT® or HCPCS code
- Revenue code
- Clinical department

- Charge code number

The charge codes also provide information to the organization on the quantity used of various supplies or the number of times a specific test was performed which can help in planning on whether or not to continue an existing service or consider removing it. Ideally the responsibility of maintaining the CDM should fall to a multi-disciplinary team with representation from the following areas. *Note that this suggested list of members on the CDM team may vary to some degree based on the organization structure, size, information system and processes.*

- Health Information
- Patient Accounts
- Information Systems
- Clinical departments that generate charges including:
    - Radiology
    - Laboratory
    - Emergency Department
    - Physical therapy
- Corporate Compliance and Physicians, as needed

How the charges are applied to a patient's account may vary by the information system, billing software and EHR features. In many cases, the charges are linked to physician orders and when the order for the service is processed, all associated charges are applied. Making sure the correct and current CPT® or HCPCS codes are associated with routine procedures like lab and radiology exams is a key responsibility for HI. Procedures where the CPT/HCPCS code(s) could vary based on the actual encounter are assigned by a coding professional. These charges in the CDM are programmed to 'look' for a CPT/HCPCS code from the HI coding/abstracting system and link them together before posting them to the patient's encounter in billing.

Making sure that the CDM is current and accurate is critical to the healthcare organization's financial viability and requires ongoing oversight. If a supply (i.e. knee joint replacement) is used to treat a patient and it's not in the CDM, the charge could be missing from the patient's account causing potential negative impact to the facility's reimbursement.

It is important for HI professionals to understand the relevance of chargemasters. Consider the following three questions:

- What are three key elements of a chargemaster list?
- What departments of the hospital should be involved in the chargemaster committee to ensure that the chargemaster is accurate and up-to-date?
- How often does an organization's chargemaster need to be reviewed and updated?

Health Information Exchange

Health information Exchanges (HIEs) are systems that allow healthcare providers to securely and appropriately, access a patient's health information electronically. This helps to improve and expedite patient care. HI is involved with HIEs as it is the patient's medical information that comes from the healchcare record they manage.

The existence and need for HIEs continues to grow and expand. HIEs help to make patient information more accessible when it's needed to deliver quality, safe, and efficient care. They also help to reduce duplication of tests and treatments and coordinate patient care more effectively between providers and healthcare organizations. Some of the benefits of a HIE include the following:

- Increases efficiency by eliminating unnecessary paperwork
- Eliminates redundant or unnecessary testing
- Improves public health reporting and monitoring

Visit HealthIT.gov to learn more information about HIEs. Consider the following two questions:

- Do you have any health information exchanges in your area?
- Have you ever been asked about allowing providers access to your health information through a HIE?